Statistics, Counting & Probability

ANSWER KEY

Section 1: Expected Value & Probability Distributions

1 A charity raffle sells $1,000$ tickets at 5 each. There is one grand prize of $1,000$ and two second prizes of 500 each. What is the expected net gain for a person purchasing one ticket?

a. **-$3.00** ✓

b. -$2.00

c. -$1.50

d. $0.00

Expected payout $= \frac{1}{1000}(1000) + \frac{2}{1000}(500) = 1 + 1 = 2$. Net expected gain $= 2 - 5 = -3$ dollars.

2 A random variable X has probability distribution: $P(X=1) = 0.2$, $P(X=2) = 0.5$, $P(X=3) = 0.3$. What is the expected value $E(X)$?

a. 2.0

b. 2.1 ✓

c. 2.2

d. 2.3

$E(X) = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1$.

3 If $E(X) = 4$ and $E(X^2) = 25$, what is the variance $\text{Var}(X)$?

a. 3

b. 5

c. 9 ✓

d. 16

$\text{Var}(X) = E(X^2) - [E(X)]^2 = 25 - 4^2 = 25 - 16 = 9$.

Section 2: Normal Distribution & Z-Scores

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4 The scores on a standardized exam are normally distributed with a mean of 500 and a standard deviation of 100. Approximately what percentage of test takers score between 400 and 600?

a. 50%

b. 68% ✓

c. 95%

d. 99.7%

Scores between 400 and 600 lie within 1 standard deviation of the mean ($\mu \pm \sigma$). By the Empirical Rule, this encompasses approximately 68% of data.

5 A student scores 84 on a test with mean 72 and standard deviation 8. What is the student's z-score?

a. 1.25

b. 1.50 ✓

c. 1.75

d. 2.00

Formula: $z = \frac{x - \mu}{\sigma} = \frac{84 - 72}{8} = \frac{12}{8} = 1.50$.

6 In a normal distribution with mean 50 and standard deviation 5, what value corresponds to a z-score of -1.6?

a. 42 ✓

b. 44

c. 46

d. 48

$x = \mu + z\sigma = 50 + (-1.6)(5) = 50 - 8 = 42$.

Section 3: Conditional Probability & Bayes' Rule

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7 Given $P(A) = 0.6, P(B) = 0.5,$ and $P(A \cap B) = 0.3$, what is the conditional probability $P(A \mid B)$?

a. 0.50

b. 0.60 ✓

c. 0.75

d. 0.30

$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.60.$

8 If two events A and B are mutually exclusive with $P(A) = 0.35$ and $P(B) = 0.45$, what is $P(A \cup B)$?

a. 0.70

b. 0.80 ✓

c. 0.90

d. 0.1575

For mutually exclusive events, $P(A \cap B) = 0$. Therefore, $P(A \cup B) = P(A) + P(B) = 0.35 + 0.45 = 0.80.$

9 A medical screening test is 95% accurate. If 2% of a population has the disease, what is the probability that a randomly chosen person has the disease AND tests positive?

a. 0.019 ✓

b. 0.020

c. 0.095

d. 0.95

$P(\text{Disease} \cap \text{Positive}) = P(\text{Disease}) \times P(\text{Positive} \mid \text{Disease}) = 0.02 \times 0.95 = 0.019.$

Section 4: Combinatorial Counting & Pigeonhole Principle

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10 In how many ways can 6 students be seated in a circle?

a. 120 ✓

b. 720

c. 60

d. 24

Circular permutations of n distinct objects is $(n-1)!$. Here $(6-1)! = 5! = 120$.

11 What is the minimum number of socks that must be drawn from a drawer containing 10 black socks and 10 white socks to guarantee getting at least one matching pair?

a. 2

b. 3 ✓

c. 4

d. 11

There are 2 colors (pigeonholes). By the Pigeonhole Principle, drawing $2 + 1 = 3$ socks guarantees at least two socks of the same color.

12 How many distinct permutations can be formed using all the letters of the word 'STATISTICS'?

a. $50,400$ ✓

b. $100,800$

c. $302,400$

d. $3,628,800$

Word length is 10 letters: 3 S's, 3 T's, 2 I's, 1 A, 1 C. Permutations $= \frac{10!}{3!\,3!\,2!} = \frac{3,628,800}{6 \times 6 \times 2} = \frac{3,628,800}{72} = 50,400$.

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